

Optimization and Machine Learning in VLSI Design Automation

Peking University, Fall 2026

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Release Date: September 8, 2026, 8:00

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Lab 0: Automatic Chip Design using OpenROAD

1 Questionnaire

Please fill out the questionnaire and provide your team information:

<https://wj.qq.com/s2/27691810/t1pf/>

2 Goal

Get familiar with the flow of Electronic Design Automation using OpenROAD.

3 Flow timeline

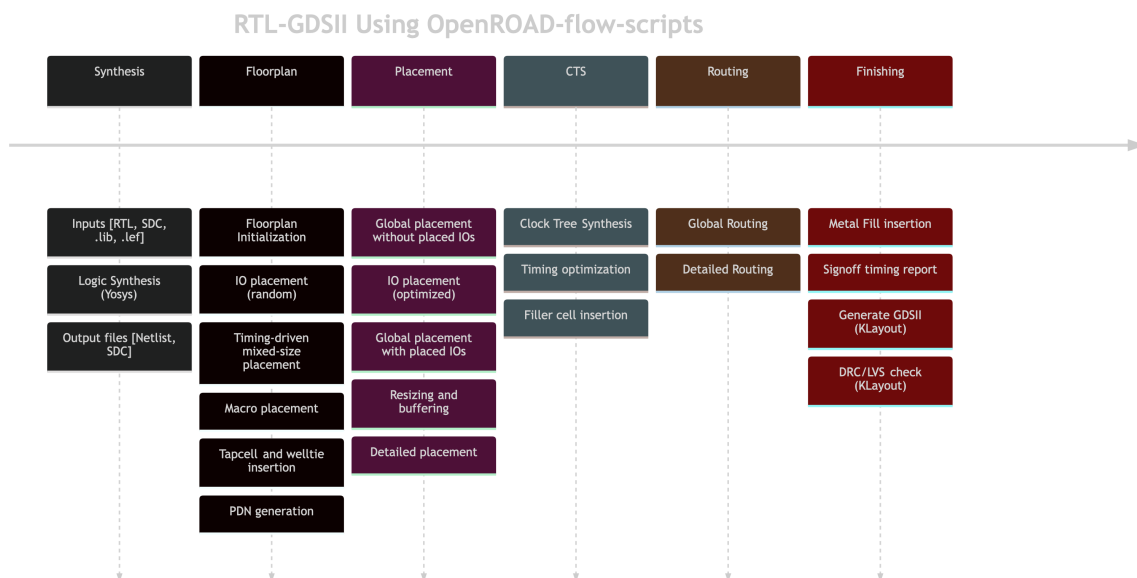


Figure 1: RTL-GDSII flow using OpenROAD-flow-scripts.

4 Case

Use the ORFS built-in **jpeg case** with the **sky130hd** platform [1].

5 Required inputs

- **RTL:** the unmodified JPEG encoder RTL supplied by ORFS.
- **Constraints:** `constraint.sdc`.
- **Technology and run configuration:** the ORFS sky130hd platform and `config.mk`.

6 Expected outputs

File names may vary by ORFS version. Use the stage prefixes in the logs:

Stage	Representative logs and results	What to inspect
Synthesis	1_1_yosys_canonicalize.log, 1_2_yosys.log, 1_synth.v	Netlist, instance counts
Floorplan	2_.log, 2_floorplan.odp	Die/core, rows, I/O, tapcells, PDN
Placement	3_.log, 3_place.odp	Placement, buffering/resizing, congestion, timing
CTS	4_.log, 4_cts.odp, 4_cts.sdc	Clock tree, timing
Routing	5_.log, 5_route.odp, 5_route_drc.rpt	Routing congestion, DRC, timing
Finishing	6_report.log, 6_final.gds	Final timing and GDS generation

7 Questions

Answer the following questions concisely and clearly.

1. **Flow stages.** List all stages in this run, from synthesis through finishing.
2. **Size, timing, and DRC.** For each stage, report the design size (total, register/sequential, and combinational-logic instance counts), timing (WNS and TNS), and DRC count.
3. **Maximum frequency.** Starting from the default 5.0 ns clock period, determine the minimum clock period that still satisfies setup timing (i.e., $WNS \geq 0$), and report the corresponding maximum frequency and DRC count. For reference, a 1.0 ns clock period may not achieve timing closure.

Hint: For setup timing,

$$RAT = T - t_{\text{setup}}, \quad WNS = RAT - AT.$$

Thus, for a tested clock period T , the limiting period can be estimated as

$$T_{\text{min}} = T - WNS = AT + t_{\text{setup}}.$$

Use the **post-routing** design only: modify the clock period in the SDC file and rerun OpenSTA. Do not rerun placement or routing.

4. **Runtime.** Report the time share of each stage:

$$\text{share}_i = \frac{t_i}{\sum_j t_j} \times 100\%.$$

Identify the likely runtime bottleneck and explain why.

5. **Errors and warnings.** Search all submitted logs for `ERROR`, `WARNING`, and `WARN`. For each nontrivial message, record its stage, cause, and whether it can be waived.
6. **Layout and critical path.** Submit a screenshot of the final layout with routing-congestion visualization enabled. Identify the most congested region and submit a view of the worst/critical timing path.

8 Submission

- Experiment logs.
- One report answering the questions in section 7.

References

- [1] “OpenROAD-flow-scripts,” GitHub.
<https://github.com/The-OpenROAD-Project/OpenROAD-flow-scripts>
- [2] “OpenROAD Flow Scripts Tutorial.”
<https://openroad-flow-scripts.readthedocs.io/en/latest/tutorials/FlowTutorial.html>
- [3] “OpenROAD,” GitHub.
<https://github.com/The-OpenROAD-Project/OpenROAD>